

Field-theoretic model of a protein with 5_2 knot topology based on the Abelian Higgs model formalism

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Abstract

In this work, we consider the application of an effective field-theoretic model to investigate the thermal stability of proteins with complex topology. The energy of the polypeptide chain is described by the discretized Hamiltonian of the Abelian Higgs model, where the geometry of the $C\alpha$ backbone is parameterized by curvature and torsion fields. The object of study is the I93M mutant of ubiquitin C-terminal hydrolase L1 (PDB: 3IRT), which possesses a 5_2 knot type. Comparative modeling of unfolding processes was performed at various exponential heating rates. It was revealed that the main unknotting pathway is realized through a direct $5_2 \rightarrow$ unknot transition, while the fraction of metastable intermediate states (including the 3_1 knot) is statistically insignificant ($<15\%$) and does not determine the transition thermodynamics. An effect of kinetic trapping in the transitional regime was discovered, manifesting as an anomalous asymmetry in the distribution of unknotting temperatures. Using weighted linear extrapolation, the equilibrium unknotting temperature of the knot in the quasistatic limit was determined. The obtained results demonstrate the effectiveness of the field-theoretic approach for predicting topological folding pathways and assessing the stability of protein structures without the computational cost of all-atom modeling.

1. Introduction

All-atom molecular dynamics (MD) methods prove to be significantly computationally expensive when modeling the time scales (microseconds and milliseconds) sufficient for the complete unfolding of a protein [5]. In this regard, the development of coarse-grained approaches based on effective field theories becomes a relevant direction. In particular, the geometry of the polypeptide chain can be described within the framework of the discretized Abelian Higgs model, which is a special case of the Ginzburg-Landau theory for a complex scalar field.

2. Theoretical Model

In this formalism, the polypeptide chain is approximated as a discrete spatial curve defined by the coordinates of the α -carbon atoms of the backbone ($C\alpha$). According to the Frenet-Serret theorem, the geometry of such a curve is uniquely determined by two

functions — curvature $\kappa(s)$ and torsion $\tau(s)$ — completely describing the shape of the curve up to spatial translations and rotations.

The basis of the model is the Hamiltonian describing the energy of the system's configuration. In the continuous limit, in terms of the field associated with the chain geometry via the Hasimoto transformation [1], the Hamiltonian is written as:

$$H = \partial_s \psi (\partial_s \psi)^* + \frac{\lambda}{2} (\psi \psi^* - \nu^2)^2, \quad (1)$$

where λ and ν are model parameters determining the coupling constant and the vacuum expectation value for the potential.

Taking into account the Hasimoto transformation:

$$\psi(s) = \kappa(s) \exp \left(i \int_0^s (\tau(s') + \alpha(s')) ds' \right), \quad (2)$$

expression (1) is rewritten through the geometric invariants — curvature $\kappa(s)$ and torsion $\tau(s)$ — as:

$$H = (\partial_s \kappa)^2 + \kappa^2 (\tau + \alpha)^2 + \frac{\lambda}{2} (\kappa^2 - \nu^2)^2. \quad (3)$$

The structure of this Hamiltonian includes:

1. **Kinetic term** $(\partial_s \kappa)^2$: ensures a smooth transition of the curvature field values between the nodes.
2. **Torsion term** $\kappa^2 (\tau + \alpha)^2$: accounts for the contribution of local torsion to the system's energy. Here, the function α sets an additional external contribution to the phase structure of the field.
3. **Potential term** $\frac{\lambda}{2} (\kappa^2 - \nu^2)^2$: represents a nonlinear "double-well" type potential (Ginzburg-Landau potential), ensuring the existence of two stable states (ordered and disordered).

The equations of motion, derived from the variation of the functional (3), admit a solution in the form of a classical soliton (kink) corresponding to structural elements:

$$\kappa = \nu \tanh(\nu \lambda (s - s_0)), \quad \tau = -\alpha. \quad (4)$$

To improve the realism of the description, the introduction of an asymmetric potential is possible:

$$V(\kappa) = \frac{\lambda}{2} (\kappa^2 - \nu_1^2)(\kappa^2 - \nu_2^2), \quad (5)$$

where $\nu_1 \neq \nu_2$ correspond to non-equivalent vacua. In this case, soliton solutions become asymmetric. Such a generalization allows describing real protein structural elements possessing different energetic stabilities.

For numerical modeling of the dynamics, the Hamiltonian is discretized. The derivative is replaced by a finite difference:

$$\partial_s \psi \rightarrow \frac{1}{a} (\psi_{i+1} - \psi_i), \quad (6)$$

where a is the characteristic step along the chain (distance between adjacent C α atoms). As a result, we obtain the discrete Hamiltonian of the system:

$$H = \sum_{i=1}^{N-1} \left[\kappa_{i+1}^2 + \kappa_i^2 - 2\kappa_{i+1}\kappa_i \cos(\tau_{i+1} + \alpha_{i+1}) + \lambda(\kappa_i + \nu_1)^2(\kappa_i - \nu_2)^2 \right]. \quad (7)$$

This model reduces the problem of modeling the all-atom protein to solving the dynamic equations for the curvature and torsion fields. This allows effectively modeling the thermodynamics and folding dynamics of the protein chain, avoiding the computational costs associated with modeling the motion of each individual atom.

3. Materials and Methods

Computational experiment protocol

Modeling is performed for the I93M mutant of ubiquitin C-terminal hydrolase L1 (PDB ID: 3IRT) [2]. To quantitatively assess the stability of the 5_2 knot, series of 100 independent unfolding simulations were conducted at eight different heating durations — 1; 5; 10; 15; 20; 25; 30; 35 million steps, with a trajectory recording interval $\Delta N = k \times N_{total}$, where $k = 0.01$.

The energy change is calculated by altering the curvature or torsion angle of a random chain residue:

$$\kappa'_i = \kappa_i + \sigma_a R, \quad \tau'_i = \tau_i + \sigma_a R, \quad (8)$$

where σ_a is the change amplitude, and R is a random number generated by a normal distribution with a mean of 0 and a variance of 1.

The acceptance of the new state was carried out according to the Glauber criterion — the configuration was accepted with a probability:

$$P = \frac{e^{-\Delta H/T}}{1 + e^{-\Delta H/T}}, \quad (9)$$

where ΔH is the change in the system's energy, and T is the current thermostat temperature.

Statistical analysis of unknotting temperatures

To quantitatively describe the topological transition $5_2 \rightarrow \text{unknot}$, the fraction of unknotted configurations $\theta(T)$ was used — the empirical cumulative distribution function of unknotting temperatures:

$$\theta(T) = \frac{1}{N} \sum_{i=1}^N \Theta(T - T_i), \quad (10)$$

where T_i is the temperature at which the 5_2 knot first disappeared in the i -th simulation, N is the total number of simulations, and $\Theta(x)$ is the Heaviside step function.

The unknotting probability density $P(T)$ is related to $\theta(T)$ by the relation:

$$P(T) = \frac{d\theta(T)}{dT}. \quad (11)$$

Due to the exponential nature of the temperature change in the simulation, the analysis is carried out in the variables $x = \log_{10} T$, where to restore the continuous profile of the conformational transition between discrete measurement values and determine the characteristic transition temperatures, the experimental data were approximated by a logistic function:

$$\theta_{\log}(T) = \frac{1}{1 + \exp[-k(\log_{10} T - \log_{10} T_{50})]}, \quad (12)$$

where T_{50} is the median unknotting temperature (at which $\theta = 0.5$), and k is the transition steepness parameter.

The approximation parameters were determined by the non-linear least squares method using the Levenberg-Marquardt algorithm. Parameter uncertainties were estimated by the bootstrap method ($B = 5000$ pseudo-samples) [6].

The characteristic transition width was defined as $\Delta T = T_{90} - T_{10}$, where the temperatures T_{10} and T_{90} are found from the conditions $\theta_{\log}(T_{10}) = 0.1$ and $\theta_{\log}(T_{90}) = 0.9$, respectively [3].

4. Results and Discussion

Temperature transition profiles for the 5_2 knot

As a result of the modeling, the dependence of the conformational transition on temperature was obtained, presented in the panels of Fig. 1. With an increase in the number of simulation steps, a systematic narrowing of the transition (ΔT) is noticeable, as well as its shift towards lower temperature values. This indicates that during rapid heating, the system does not have time to overcome the unknotting energy barriers, which causes kinetic delay and an overestimated assessment of the knot's thermodynamic stability [7, 8].

To assess the statistical significance of the 3_1 knot population at each temperature point, 95% confidence intervals were constructed using Wilson's method. Across the entire studied temperature range, the upper bound of the CI reaches 18.6% only in the case of 25 million steps, remaining below 15% in all other cases. An exact binomial test with $H_0 : p_o = 0.02$ and Bonferroni correction ($\alpha' = \alpha/M$, where M is the number of temperature points, $\alpha = 0.05$) does not allow rejecting H_0 . The conservativeness of the correction is guaranteed by Boole's inequality: $P(\bigcup_{j=1}^M E_j) \leq \sum_{j=1}^M P(E_j) \leq M \cdot (\alpha/M) = \alpha$, which limits the family-wise error rate across the entire series of tests. A state was considered thermodynamically insignificant if the upper confidence bound remained below the threshold $\epsilon = 0.15$ [3].

Thus, the 3_1 knot is not a thermodynamically significant intermediate state, remaining within the limits of statistical noise. Consequently, the system can be approximated as quasi-two-state ($5_2 \leftrightarrow \text{unknot}$), which justifies the application of a two-parameter logistic model for determining melting temperatures.

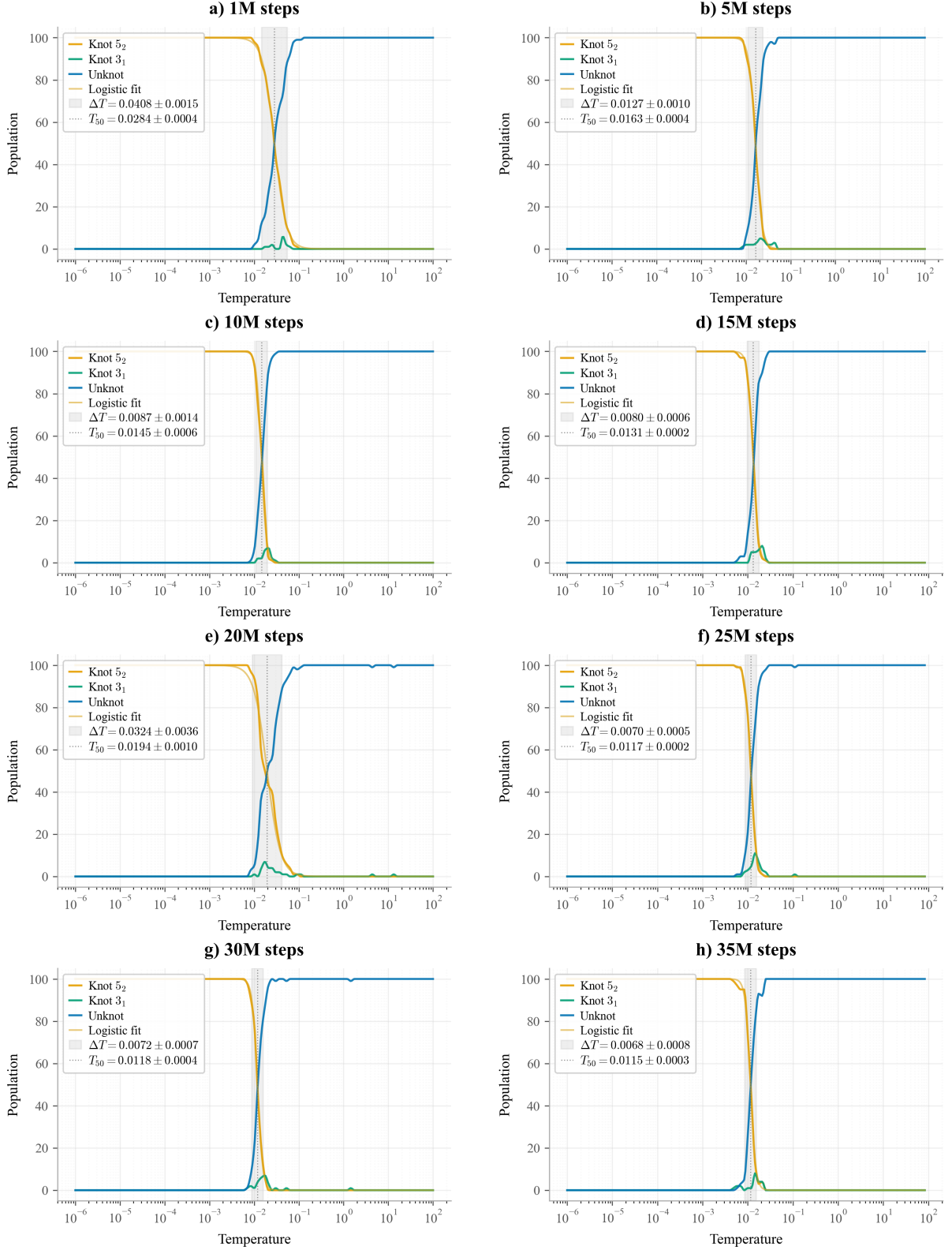


Figure 1: Dependence of the population of topological states on temperature at various heating durations. The values of the T_{50} and ΔT parameters were obtained by non-linear approximation of the population dependencies by a logistic function.

The transition parameters obtained by logistic function approximation are presented in Table 1.

Table 1: Summary metrics of transition parameters for each simulation duration. Standard errors for each parameter were determined by the bootstrap method ($B = 5000$).

Regime (steps)	$T_{10} \pm \delta$	$T_{90} \pm \delta$	$\Delta T \pm \delta$	$T_{50} \pm \delta$
$1.0 \cdot 10^6$	0.0146 ± 0.0004	0.0554 ± 0.0013	0.0408 ± 0.0015	0.0284 ± 0.0004
$5.0 \cdot 10^6$	0.0112 ± 0.0004	0.0238 ± 0.0009	0.0127 ± 0.0010	0.0163 ± 0.0004
$10.0 \cdot 10^6$	0.0108 ± 0.0008	0.0195 ± 0.0011	0.0087 ± 0.0014	0.0145 ± 0.0006
$15.0 \cdot 10^6$	0.0097 ± 0.0003	0.0178 ± 0.0005	0.0080 ± 0.0006	0.0131 ± 0.0002
$20.0 \cdot 10^6$	0.0091 ± 0.0009	0.0416 ± 0.0033	0.0324 ± 0.0036	0.0194 ± 0.0010
$25.0 \cdot 10^6$	0.0087 ± 0.0003	0.0157 ± 0.0004	0.0070 ± 0.0005	0.0117 ± 0.0002
$30.0 \cdot 10^6$	0.0088 ± 0.0005	0.0160 ± 0.0005	0.0072 ± 0.0007	0.0118 ± 0.0004
$35.0 \cdot 10^6$	0.0086 ± 0.0003	0.0154 ± 0.0006	0.0068 ± 0.0008	0.0115 ± 0.0003

When analyzing the temperature profiles, a significant anomaly was revealed in the 20 million steps regime (Fig. 1e, Table 1): the transition width $\Delta T = 0.0324$ exceeds the values for adjacent regimes (15 and 25 million) by more than 4 times; the temperature $T_{50} = 0.0194$ demonstrates a deviation from the monotonic trend. This indicates the presence of a non-trivial statistical distribution in this regime, which cannot be characterized solely by median values and requires the analysis of full probability distribution functions [3].

Statistical structure of the transitional regime

To study in detail the statistical properties of the 5_2 knot unknotting process, histograms of the distribution of individual unknotting temperatures $P(T)$ were constructed for the 20 million steps regime and the adjacent 15 and 25 million steps simulation regimes (Fig. 2).

The distribution for the 20 million steps regime (Fig. 2b) demonstrates an asymmetric shape with a heavy right tail up to $T > 0.1$, which is an order of magnitude higher than the characteristic values for adjacent regimes. The standard deviation $\sigma \approx 0.0175$ is 5 times greater than the values for the 15 million ($\sigma \approx 0.0037$) and 25 million ($\sigma \approx 0.0032$) steps regimes.

Such behavior indicates a low-cooperative nature of the transition: a part of the trajectories overcomes the topological barrier at low temperatures ($T \sim 0.012$), while the other part gets trapped in metastable states, requiring significant overheating ($T > 0.05$) to complete the transition [4]. This explains the observed discrepancy between the median of the distribution ($T_{50}^{CDF} = 0.0194 \pm 0.010$, Fig. 1e) and its mathematical expectation (the position of the center of the probability density function, obtained by differentiating the logistic CDF, $T_{50}^{fit} = 0.0227 \pm 0.0017$, Fig. 2b), which is characteristic of asymmetric distributions with heavy tails [3].

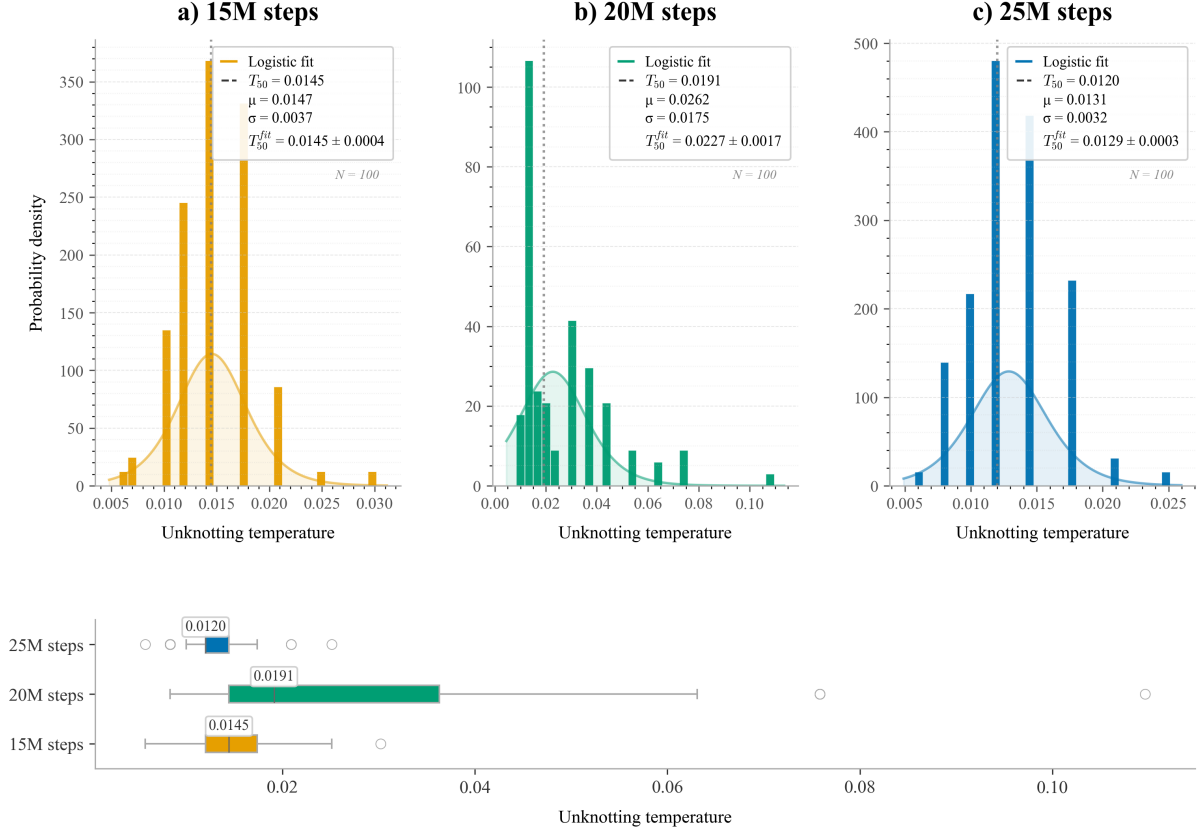


Figure 2: Analysis of the unknotting temperature distribution of the 5_2 knot for $N = 100$ simulations at 15, 20, and 25 million steps. Upper panel: probability density histograms. Solid lines represent the derivative of the logistic approximation of the cumulative distribution function (CDF, Fig. 1). The gaps between the bars and their anomalous height are due to the finite resolution of trajectory recording (100 frames) and the exponential nature of the heating, which leads to the discretization of the distribution. Lower panel: box plots indicating the medians.

Thermodynamic limit and equilibrium unknotting temperature

To determine the true thermodynamic stability of the 5_2 knot, an extrapolation of the observed transition temperatures T_{50} to the limit of infinitely slow heating ($N \rightarrow \infty$) was performed. The inverse of the simulation duration $x = 1/N$ was used as the regression variable: $x \rightarrow 0$ as $N \rightarrow \infty$.

Taking into account the anomalous variance of the 20 million steps point, the weighted least squares (WLS) method was applied for the extrapolation. The weights w_i for each point were defined as the inverse squares of the standard errors: $w_i = 1/\sigma_i^2$.

The points corresponding to 30 and 35 million steps were excluded from the WLS regression model, since their values presumably reach a thermodynamic plateau (cessation of dependence on heating time) at $T_{50} \approx 0.0117$ (Fig. 1g,h). Extrapolation to the limit $N \rightarrow \infty$ yields the value $T_{eq} = 0.0117 \pm 0.0001$, which coincides with high accuracy with the average value of the experimental plateau (30-35 million). Thus, the system indeed reaches a quasi-equilibrium state. The 20 million steps point is an anomaly associated with trapping in a local energetic minimum, rather than a thermodynamic property of the system [3].

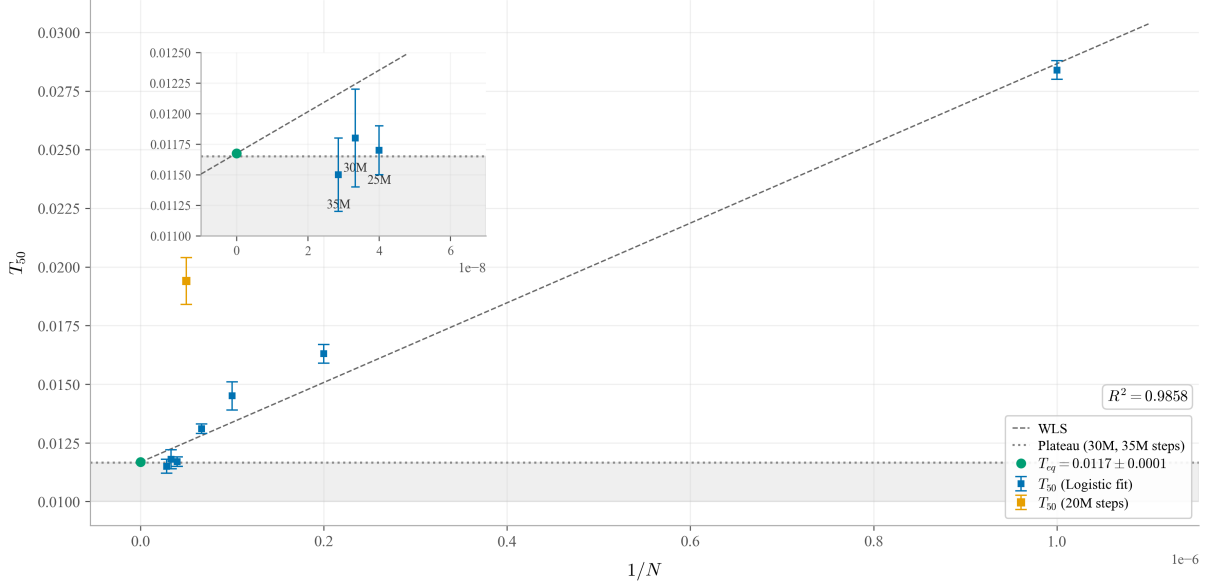


Figure 3: Extrapolation of the melting temperature of the 5_2 knot to the thermodynamic limit. The 30 and 35 million points are excluded from the regression model and lie on the thermodynamic plateau (dashed line at $T_{50} \approx 0.0117$).

5. Conclusion

Using the example of the I93M mutant of ubiquitin C-terminal hydrolase L1, we demonstrated the applicability of the considered field-theoretic model, based on the formalism of the Abelian Higgs model, for describing topological transitions in protein structures. This approach allows effectively investigating the thermodynamics and kinetics of protein unfolding, overcoming the limitations of classical molecular dynamics in terms of time scales.

It was established that the unknotting of the 5_2 knot occurs predominantly via the direct pathway $5_2 \rightarrow \text{unknot}$. Intermediate topological states (in particular, the 3_1 knot) do not affect the global thermodynamics of the transition — the system is quasi-two-state.

It is demonstrated that increasing the simulation duration leads to a systematic kinetic shift of the transition temperature. By the method of weighted linear extrapolation to the limit of infinitely slow heating ($N \rightarrow \infty$), the equilibrium unknotting temperature $T_{eq} \approx 0.0117$ (in the dimensionless units of the model) was determined. For this model, the effect of kinetic trapping (20 million steps regime) was discovered and statistically substantiated, in which the distribution of unknotting temperatures acquires an asymmetric character with a heavy tail due to the trapping of a part of the trajectories in metastable local minima.

Further development of the approach may include accounting for specific amino acid interactions, the influence of the solvent, and entropic barriers, which will improve the accuracy of predicting folding kinetics for a wide class of proteins with deep topology.

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